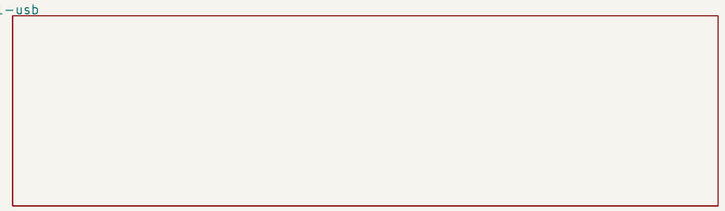
OPENKEYS R0.5 – ONE USB-C + NFC / BLUE 16x2 LCD
Device PIN mandatory / TEN DECIMAL KEYS / NO FINGERPRINT
PCB candidate; qualification required before secret provisioning

01-usb



File: 01-usb.kicad_sch

02-power



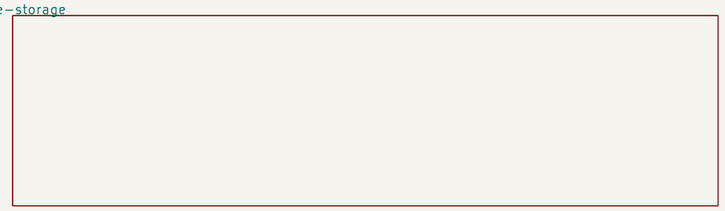
File: 02-power.kicad_sch

03-mcu



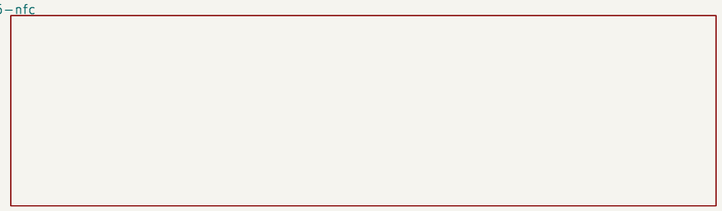
File: 03-mcu.kicad_sch

04-se-storage



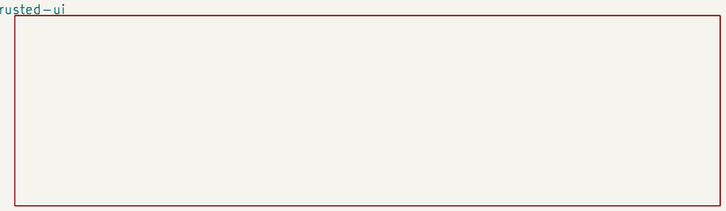
File: 04-se-storage.kicad_sch

05-nfc



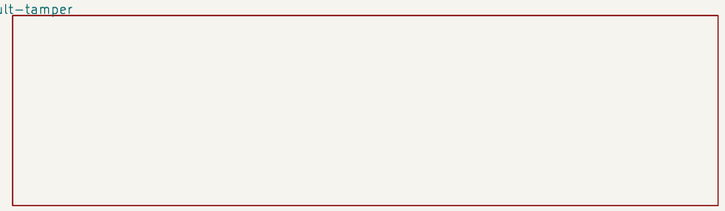
File: 05-nfc.kicad_sch

06-trusted-ui



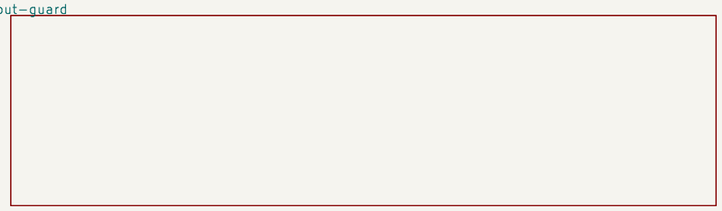
File: 06-trusted-ui.kicad_sch

07-fault-tamper



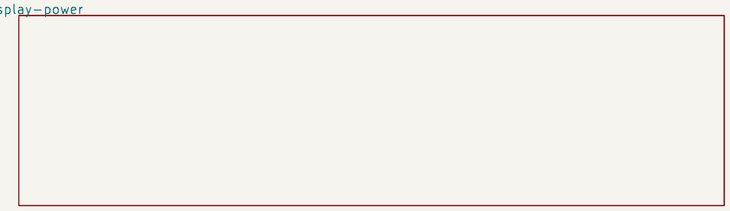
File: 07-fault-tamper.kicad_sch

08-input-guard



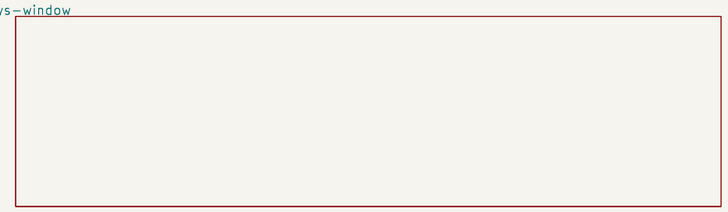
File: 08-input-guard.kicad_sch

09-display-power



File: 09-display-power.kicad_sch

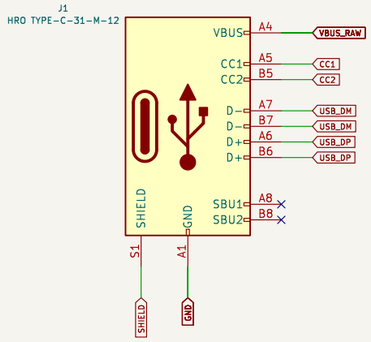
10-sys-window

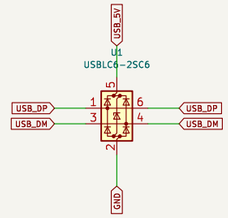


File: 10-sys-window.kicad_sch

Jay-Void-BA
Sheet: /
File: OpenKeys.kicad_sch
Title: OpenKeys R0.5 – Engineering review
Size: A3 | Date: 2026-11-13 | **Rev: 0.5**
KiCad E.D.A. 9.0.2 | Id: 1/11

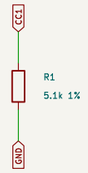
One USB-C receptacle — USB 2.0 FS data and 5 V charging

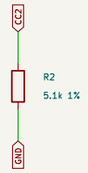


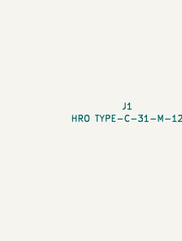


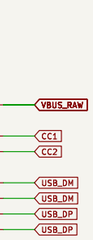




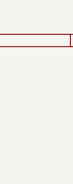




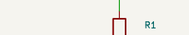


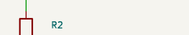












5.1k Rd on each CC pin: sink only. No PD negotiation, no 9 V request.
Apple 20 W adapter uses 5 V through a standards-compliant USB-C cable.
100mA hardware input limit: 5 V x 0.1 A = 0.50 W nominal input budget.
Input guard U14 is independent of MCU: see sheet08. Analog timing/thermal protection not yet qualified.

ENGINEERING BOARD CANDIDATE — qualify before use

Jay-Void-BA

Sheet: /01-usb/
File: 01-usb.kicad_sch

Title: OpenKeys R0.5 — 01-usb

Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 2/11

Low-rate charging, protected 500 mAh pack and 3.3 V supply

#FLG3
PWR_FLAG

VBUS_FUSED

VBUS_RAW

F1
250mA hold – qualify I2t

VBUS_FUSED

U2
BQ25185DLHR

SYS	1	SYS	
BAT	2	BAT	
CHG_STAT2_N	3	STAT2	
CHARGE_DISABLE	4	CE_N	
GND	5	GND	
BAT_NTC	6	TS_MR	
	7	ILIM_VSET	CHG_ILIM_VSET
	8	ISET	CHG_ISET
	9	STAT1	CHG_STAT1_N
	10	IN	USB_5V
	11	EP	GND

J2
Adafruit 1578 protected 1S 500mAh

BAT	1
GND	2

CHG_ILIM_VSET

R6
24k 1%

GND

CHG_ISET

R7
3.00k 1%

GND

SYS

R8
100k

CHARGE_DISABLE

CHARGE_DISABLE

CHARGE_ENABLE 1

Q2
2N7002

GND

CHARGE_ENABLE

R9
100k

GND

J6
103AT-2 external battery thermistor

BAT_NTC	1
GND	2

SW_L1

L1
1.5uH Isat>=1.5A

SW_L2

SYS

R10
0R

REG_VINA

#FLG4
PWR_FLAG

REG_VINA

U3
TPS63031DSK (3.3 V)

SYS	5	VIN	
REG_VINA	8	VINA	
REG_EN	6	EN	
SYS	7	PS/SYNC	
SW_L1	4	L1	
SW_L2	2	L2	
GND	9	GND	
GND	3	PGND	
	1	VOUT	+3V3
	10	FB	+3V3

REG_EN 2

SW1
OFF / ON

1 SYS
3 GND

USB_5V

C2
1uF 10V

GND

BAT

C3
10uF 6.3V

GND

SYS

C4
10uF 10V

GND

SYS

C5
10uF 10V

GND

REG_VINA

C6
100nF

GND

+3V3

C7
10uF 6.3V

GND

+3V3

C8
10uF 6.3V

GND

Fixed100mA input; charge target100mA=0.2C for500mAh.
Input DPM gives system load priority, so cell charge can be slower.
CE defaults disabled. Public charging service never unlocks secrets.
Suspend: stop charge/backlight; qualify total VBUS current <=2.5mA.
Real pack PCM + attached thermistor required. No fast charge.

ENGINEERING BOARD CANDIDATE – qualify before use

Jay-Void-BA

Sheet: /02-power/
File: 02-power.kicad_sch

Title: OpenKeys R0.5 – 02-power

Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 3/11

STM32H573RIT6 — STiRoT and TrustZone protected I/O

C9
100nF

C10
100nF

C11
100nF

C12
100nF

C13
100nF

C14
4.7uF

C15
2.2uF low ESR

C16
2.2uF low ESR

C17
100nF

C18
1uF

U4
STM32H573RIT6

		VBAT	1
		VDD	19
		VDD	32
		VDD	48
		VDD	64
		VDDA	43
NRST	7	NRST	
BOOT0	60	BOOT0	
	5	PH0	
	6	PH1	
	54	PD2	
USB_GUARD_FAULT_N	8	PC0	
NRST	9	PC1	
FLASH_MISO	10	PC2	
LCD_RELEASE	11	PC3	
LCD_ARM_CLK	24	PC4	
VBUS_PRESENT_N	25	PC5	
BTN5_N	37	PC6	
BTN6_N	38	PC7	
SE_ENABLE_REQUEST	39	PC8	
LCD_HOST_SDA	40	PC9	
BTN8_N	51	PC10	
BTN9_N	52	PC11	
BTN0_N	53	PC12	
CASE_CLOSED	2	PC13	
LSE_IN	3	PC14	
LSE_OUT	4	PC15	
VCAP1	30	VCAP	
VCAP2	62	VCAP	
	14	PA0	
	15	PA1	
CHG_STAT1_N	16	PA2	
CHG_STAT2_N	17	PA3	
NFC_CS_N	20	PA4	
NFC_SCK	21	PA5	
NFC_MISO	22	PA6	
NFC_MOSI	23	PA7	
LCD_HOST_SCL	41	PA8	
	42	PA9	
	43	PA10	
USB_DM	44	PA11	
USB_DP	45	PA12	
SWDIO	46	PA13(JTMS	
SWCLK	49	PA14(JTCK	
BTN7_N	50	PA15(JTDI)	
BTN1_N	26	PB0	
BTN2_N	27	PB1	
BTN3_N	28	PB2	
SWO	55	PB3(JTDO	
CHARGE_ENABLE	56	PB4(NJTRST)	
BTN4_N	57	PB5	
SE_SCL	58	PB6	
SE_SDA	59	PB7	
	61	PB8	
NFC_IRQ_N	29	PB10	
FLASH_CS_N	33	PB12	
FLASH_SCK	34	PB13	
	35	PB14	
FLASH_MOSI	36	PB15	
	18	VSS	
	12	VSSA	

FB1
600R@100MHz

#FLG6
PWR_FLAG

R11
10k

C19
10nF — qualify reset timing

R12
10k

Y1
ABS07-32768KHZ-6-T

C20
9pF C0G — tune

C21
9pF C0G — tune

J3
Internal SWD — DNP production

+3V3	1	2	SWDIO
GND	3	4	SWCLK
GND	5	6	SWO
	7	8	
GND	9	10	NRST

STiRoT must verify protected firmware. TrustZone/GTZC must deny nonsecure access to SE, LCD, ten keys, timers, DMA and key memory.
No raw cryptography or arbitrary display commands at secure gateway.
SE_EN is a fault interlock, not proof of a correct PIN.
Separate2.2uF on VCAP1/2; no external load on VCAP.

ENGINEERING BOARD CANDIDATE — qualify before use

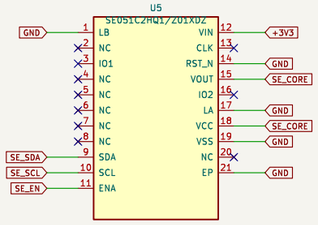
Jay-Void-BA

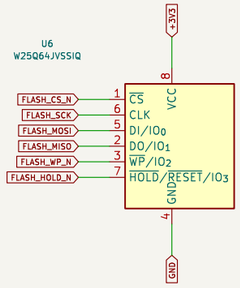
Sheet: /03-mcu/
File: 03-mcu.kicad_sch

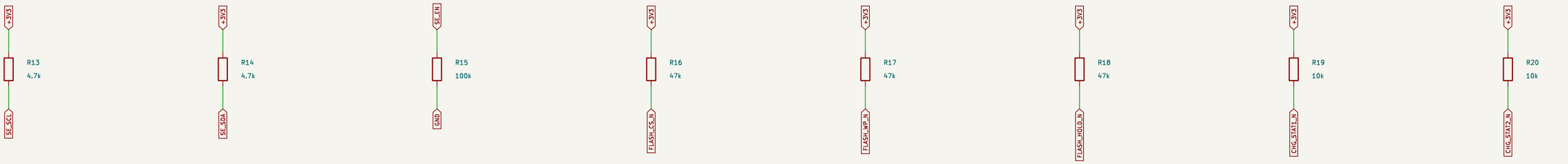
Title: OpenKeys R0.5 — 03-mcu

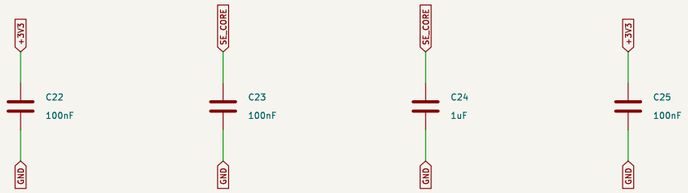
Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 4/11

Secure element, protected keys and encrypted records









Private operations: valid local six-digit PIN grant + per-request confirmation.
No biometric path. SE-native PIN ACLs and durable retries must be qualified; Root owns keypad.
Precharge attempts durably; uncertain writes deny. AEAD does not alone prevent rollback.
SE certification does not certify MCU, PCB, PIN execution path or complete product.

ENGINEERING BOARD CANDIDATE — qualify before use

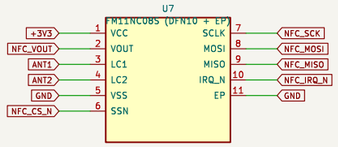
Jay-Void-BA

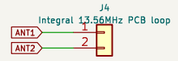
Sheet: /04-se-storage/
File: 04-se-storage.kicad_sch

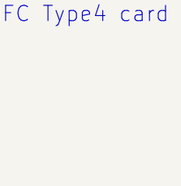
Title: OpenKeys R0.5 — 04-se-storage

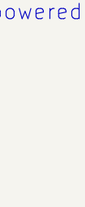
Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 5/11

NFC Type4 card interface — battery-powered local PIN UI









NFC VOUT isolated from SYS/BAT/+3V3. No assumed RF-powered full device.
LCD/PIN/CTAP NFC require charged battery and device ON.
RF region: tuned coil only, no cell, metal shield or ground planes.
Mobile management needs authenticated APDUs and native OS integration.
Phone biometrics or a tap never replace device PIN verification.

ENGINEERING BOARD CANDIDATE — qualify before use

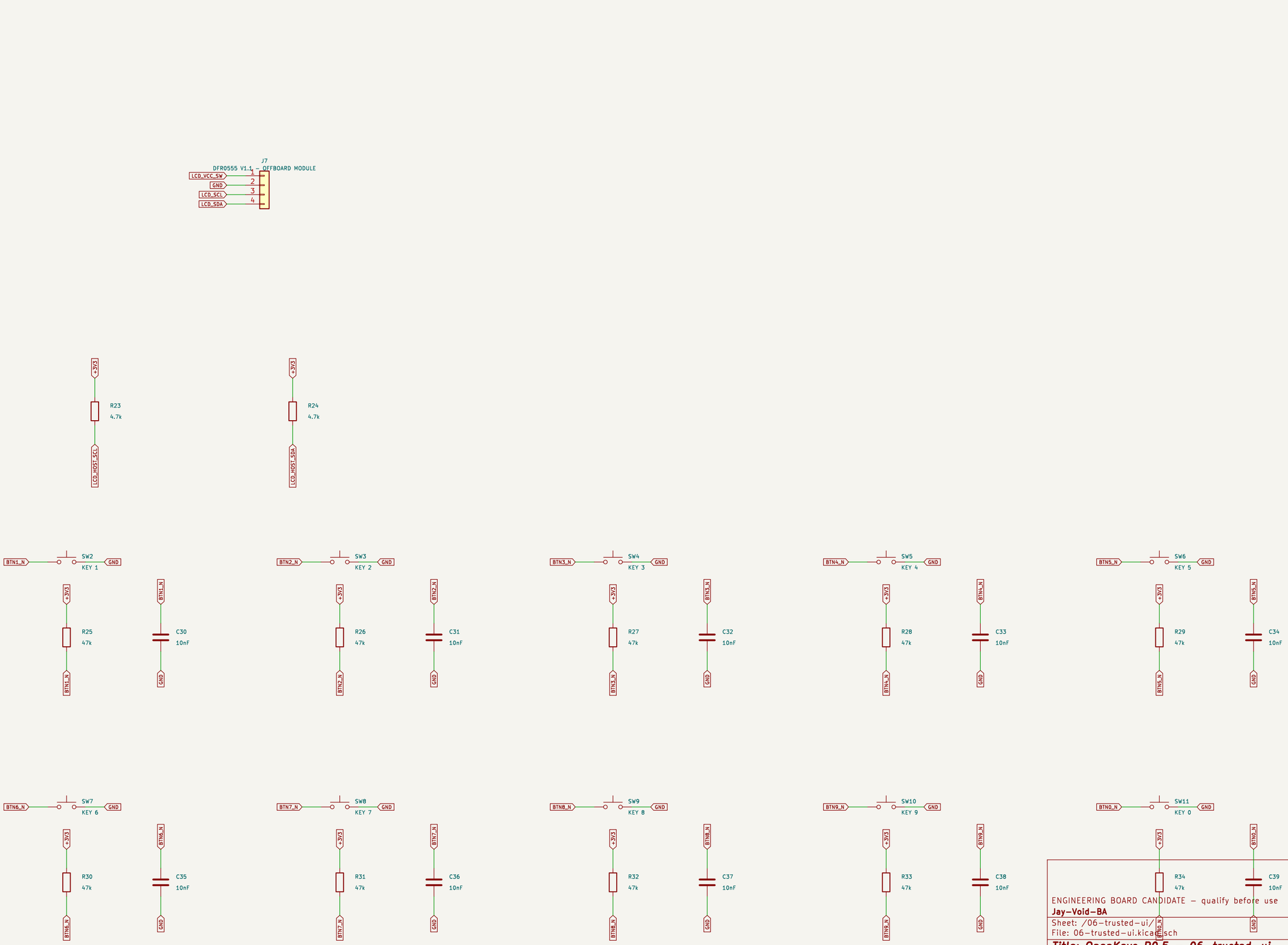
Jay-Void-BA

Sheet: /05-nfc/
File: 05-nfc.kicad_sch

Title: OpenKeys R0.5 — 05-nfc

Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 6/11

DFR0555 blue 16x2 LCD and ten independent PIN keys



ENGLISH 16x2, blue background / white text. No fingerprint parts.
PIN:6 decimal digits, masked. Hold9=erase; hold0=cancel.
Review:2=previous,8=next,5=confirm,0=cancel. New release required.
Private operations require a valid paired3Os local PIN grant and exact review.
No PIN from USB/NFC/desktop. No factory/default unlock PIN.

ENGINEERING BOARD CANDIDATE – qualify before use

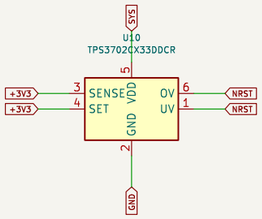
Jay-Void-BA

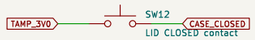
Sheet: /06-trusted-ui/
File: 06-trusted-ui.kicad_sch

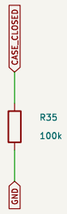
Title: *OpenKeys R0.5 – 06-trusted-ui*

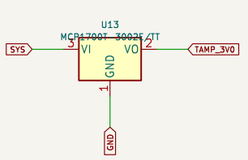
Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 7/11

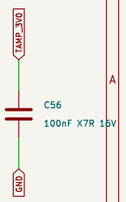
Independent reset, battery–backed case tamper and SE interlock

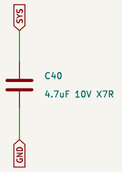


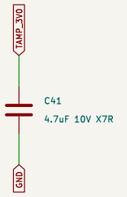


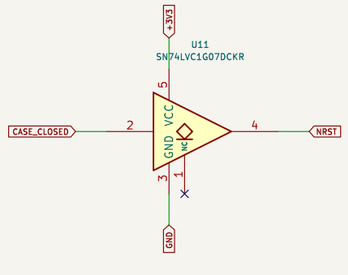


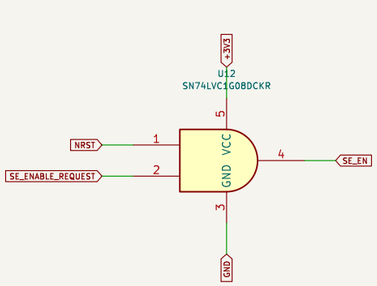




















Reset invalidates volatile grants; reboot must verify firmware and require fresh device PIN.
PC13=case input; PC1/TAMP_IN3 observes reset/fault line.
Internal BOR/PVD, clock checks and persistent precharged attempts also required.
Production debug lifecycle closed; no populated external debug connector.
VBAT+LSE must retain tamper with main OFF; full backup loss must lock, not initialize.
Measure reset/ENA latency, leakage and injection before security release.

ENGINEERING BOARD CANDIDATE – qualify before use

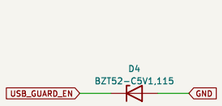
Jay–Void–BA

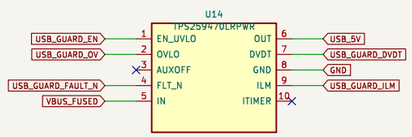
Sheet: /07–fault–tamper/
File: 07–fault–tamper.kicad_sch

Title: OpenKeys R0.5 – 07–fault–tamper

Size: A3	Date: 2026–11–13	**Rev: 0.5–board–review**
KiCad E.D.A. 9.0.2		Id: 8/11

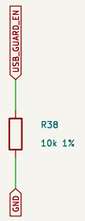
Independent VBUS overvoltage cutoff, reverse blocking and limited inrush

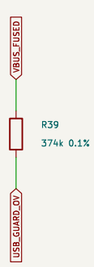


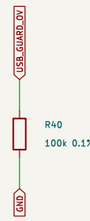


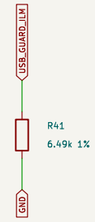


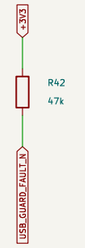


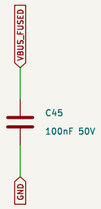


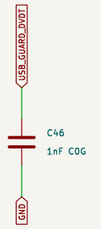












OVLO nominal1.2V x(1+374k/100k)=5.688V. Component/leakage corners in protection-calculations.json.
UVLO divider22.1k/10k: about3.85V nominal. EN is separately clamped, never raw VBUS.
6.49k ILM: about0.51A protection threshold; BQ charger still sets100mA input budget.
1nF DVDT: about2V/ms, about2.5ms ramp to5V; measure actual inrush with complete load.
F1 -> input TVS -> eFuse -> output TVS/ESD reference/charger/sense.
Independent cutoff is a circuit provision; response overshoot, CC/data faults and thermal/surge tests remain BLOCKED.

ENGINEERING BOARD CANDIDATE - qualify before use

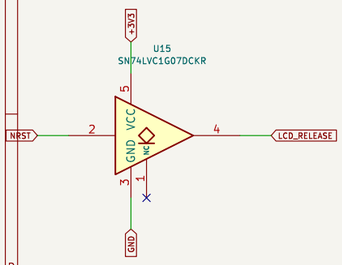
Jay-Void-BA

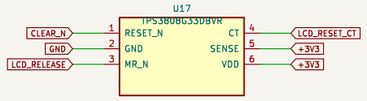
Sheet: /08-input-guard/
File: 08-input-guard.kicad_sch

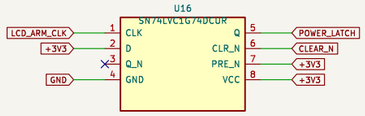
Title: OpenKeys R0.5 - 08-input-guard

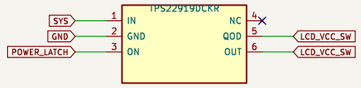
Size: A3	Date: 2026-11-13	**Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2		Id: 9/11

Latched LCD power removal; isolated bus; no resetless retained text

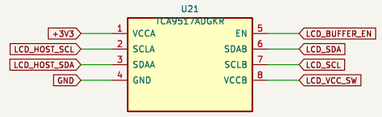




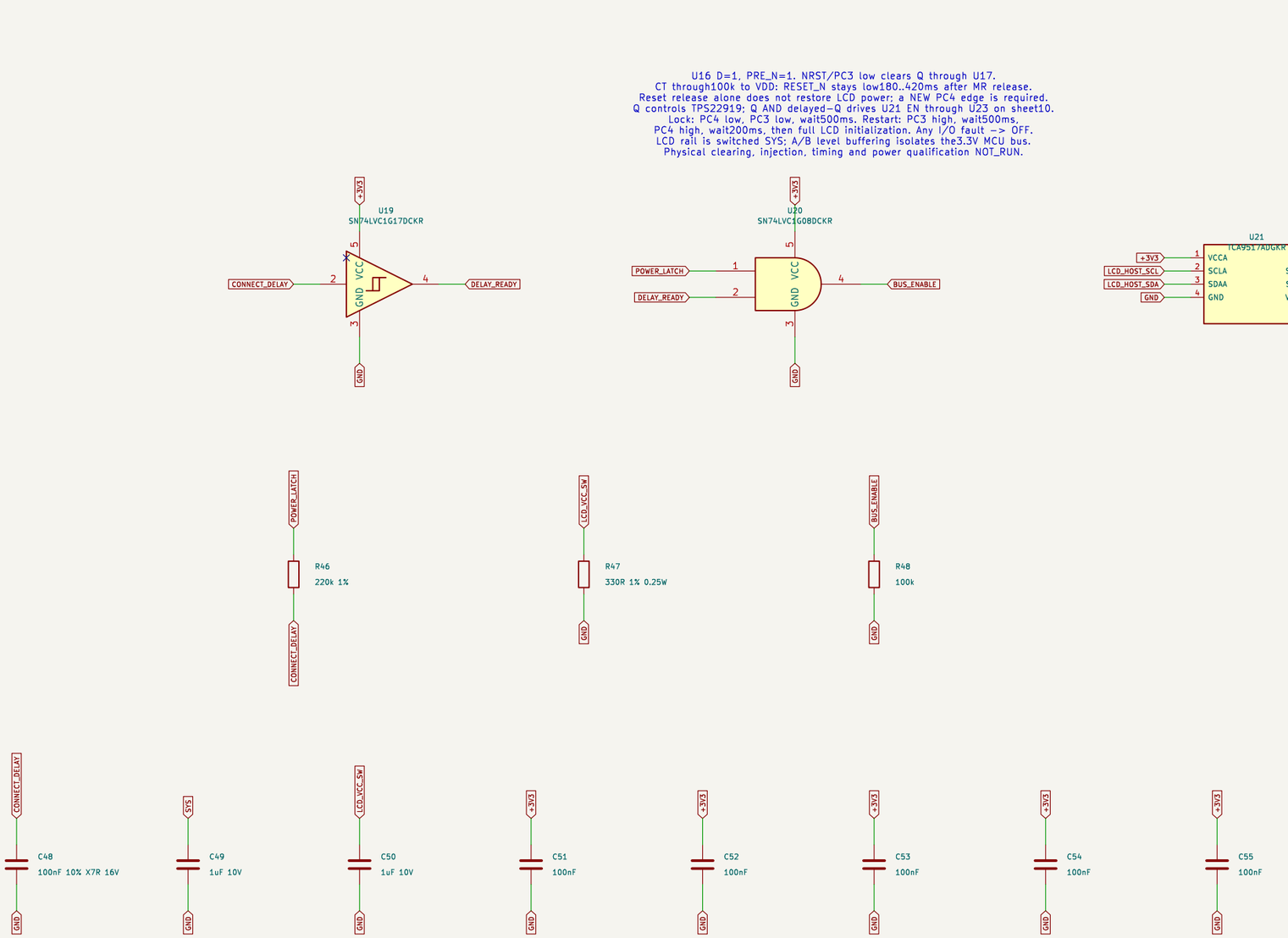




U16 D=1, PRE_N=1. NRST/PC3 low clears Q through U17.
CT through100k to VDD: RESET_N stays low180..420ms after MR release.
Reset release alone does not restore LCD power; a NEW PC4 edge is required.
Q controls TPS22919; Q AND delayed-Q drives U21 EN through U23 on sheet10.
Lock: PC4 low, PC3 low, wait500ms. Restart: PC3 high, wait500ms,
PC4 high, wait200ms, then full LCD initialization. Any I/O fault -> OFF.
LCD rail is switched SYS; A/B level buffering isolates the3.3V MCU bus.
Physical clearing, injection, timing and power qualification NOT_RUN.

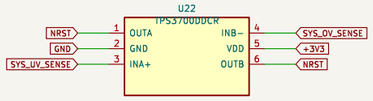


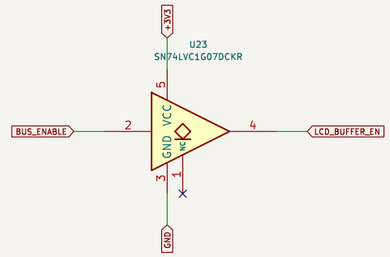




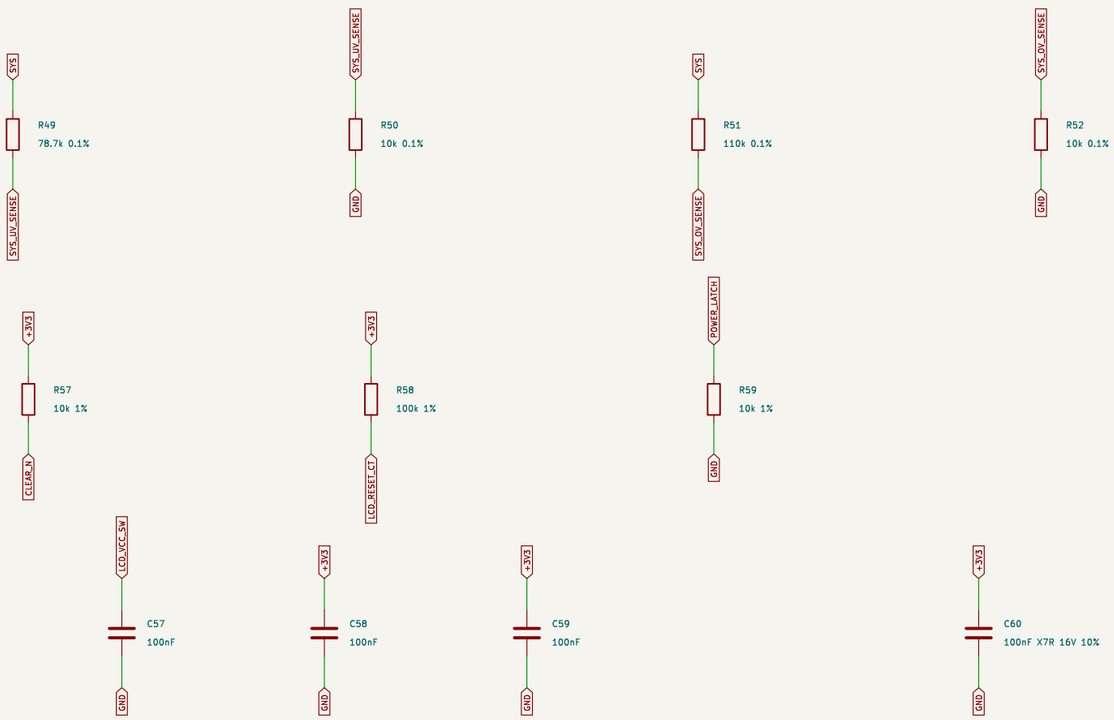
ENGINEERING BOARD CANDIDATE — qualify before use
Jay-Void-BA
Sheet: /09-display-power/
File: 09-display-power.kicad_sch
Title: OpenKeys R0.5 — 09-display-power
Size: A3 | Date: 2026-11-13 | **Rev: 0.5-board-review**
KiCad E.D.A. 9.0.2 | Id: 10/11

SYS voltage window and LCD level–buffer fail–closed enable





TPS3700 reference: positive396..404mV; negative387..400mV.
Include0.1percent divider errors and +/–25nA input bias.
Static SYS UV fall min3.4246V; OV rise max4.8596V.
With ASSUMED switch+harness drop<=50mV: LCD>=3.3746V.
Main3.3V fixed PWM:3.267..3.333V; no ripple/transient guarantee.
TCA sideA=main3.3V; sideB=switched SYS. Off I2C must stay high–Z.
EN high follows switched VCCB through U23, R53/R54.
LCD total C<=100uF and aggregate off injection<=20uA MUST be measured.
Hardware hold180ms minus ASSUMED turn–off<=5ms permits discharge.
Services stay locked during startup, undervoltage and all faults.
Fast pulses, backfeed, input budget, temperature and RF remain unqualified.



ENGINEERING BOARD CANDIDATE – qualify before use
Jay–Void–BA

Sheet: /10–sys–window/		
File: 10–sys–window.kicad_sch		
Title: OpenKeys R0.5 – 10–sys–window		
Size: A3	Date: 2026–11–13	**Rev: 0.5–board–review**
KiCad E.D.A. 9.0.2		Id: 11/11